

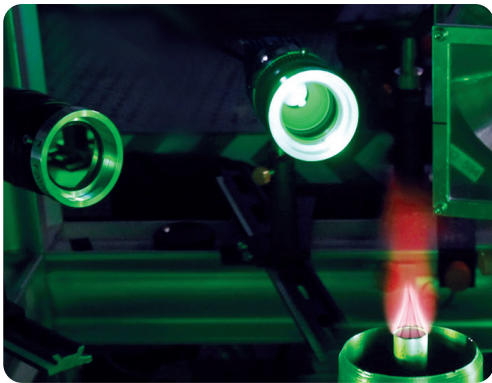
Singapore, November 2026

Dear conference participant,

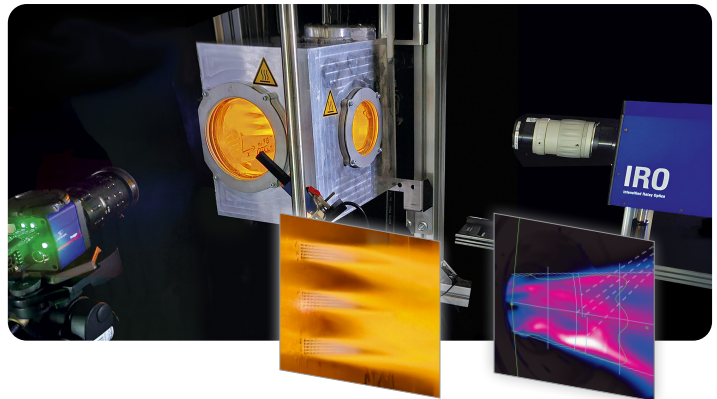
Welcome to the **2nd SSCI Annual Combustion Meeting** in the vibrant city of Singapore.

LaVision is proud to support the combustion community with advanced imaging solutions since more than 35 years, and we are pleased to present our latest product innovations to you at the booth from our valued partner Analytical Technologies:

Hydrogen and Ammonia imaging: Hydrogen is increasingly important as a carbon-free fuel for heat, vehicles, energy storage, and long-range transport in industrial applications. Ammonia, traditionally used in agriculture and industry, is now also a promising alternative fuel for large maritime engines.



*Imaging of a laminar ammonia flame
Courtesy of TU Darmstadt*

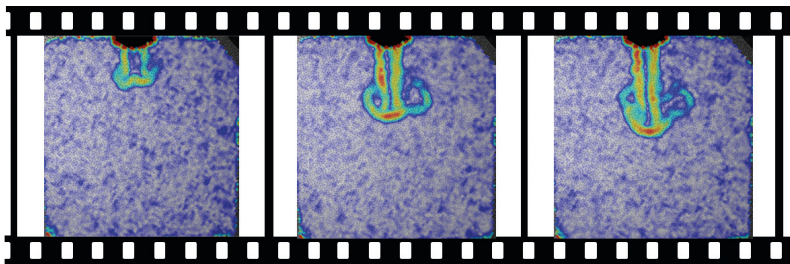


Visualization of hydrogen combustion

LaVision's **FlameMaster** system family is designed to find new concepts for the realization of more efficient and cleaner combustion devices. These (laser) imaging systems are designed for multi-parameter measurements in a large variety of flames with high spatial and temporal resolution.

LaVision's **FluidMaster BOS** system allows the visualization of both: non-reacting and reacting gas flows, such as hydrogen or ammonia, without the need of added seeding.

Visualization of gas flows: LaVision's **FlowBOS** imaging systems visualize gas flows in front of a background pattern without particle contamination. The live system detects smallest optical changes of the flow (aka Schlieren) either due to temperature gradients in thermal flows or gradients from mixing of different gases.



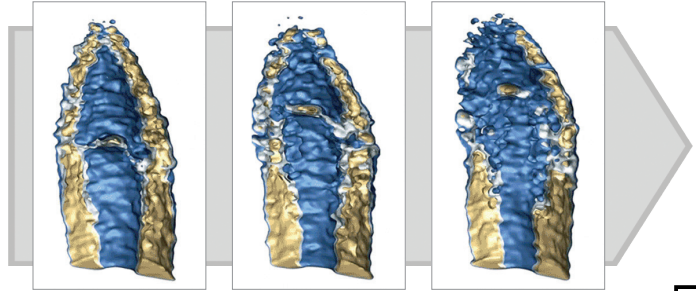
BOS imaging of ammonia injection, courtesy of IFKM (KIT, Germany)



FlowBOS camera with tablet

The **BOS** technique visualizes the flow structure of the penetrating gas plume. The subsequent quantification of parameters like cone angle, penetration depth or perimeter enables a better understanding of the mixing process.

Volumetric imaging of combustion species: Our **FlameMaster 3D-LIF** imaging systems support instantaneous 3D-LIF imaging of OH in flames for turbulent combustion analysis. Multiple intensified cameras with image doublers are used for the tomographic reconstruction of the 3D-LIF signal.



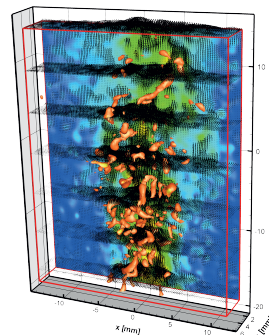
Instantaneous 3D-LIF imaging of OH in jet flames using tomographic reconstruction

High-speed intensifier with ultra-fast shutter: The modular design of LaVision's high-speed intensifier (IRO) adds maximum sensitivity to a wide range of camera models. The **HighSpeed IRO X** is available with a single-stage amplifier (MCP) and as a 2-stage system (MCP with booster) optimized for measurements at highest frame rates.



High-resolution camera sensitive for UV: The **Imager CS3-8-UV** camera assures low noise at excellent sensitivity and provides UV sensitivity down to 200 nm. With 8 MP resolution, global shutter and 2x2 binning capability, the camera provides outstanding features for imaging applications.

Compact 3D multi-sensor camera for full volumetric flow measurements: The **MiniShaker** is an aligned multi-sensor system in a compact housing for quick and easy volumetric flow measurements. Integrated into LaVision's well known DaVis software platform, full 3D flow fields are readily obtained and quantified means of the powerful and proven Shake-the-Box (4D-PTV, STB) and Tomographic PIV packages. Additionally, a unique, integrated uncertainty estimation tool for 3D flow (STB) data is available.



We are happy to help you solving your measurement tasks with our intelligent imaging systems!

Your LaVision Team

